CodeAlpha Internship

Python Programming

Internship Report

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Found via LinkedIn - Online Internship Program

Introduction

This report summarises the work completed during the Python Programming internship at CodeAlpha. The internship was discovered through a LinkedIn online internship posting and provided practical, project-based experience in core Python concepts.

Three distinct tasks were undertaken, each targeting different aspects of Python development:

- Task 1: Hangman Game (logic, loops, string manipulation)
- Task 2: Stock Portfolio Tracker (dictionaries, I/O, file handling)
- Task 3: JPG File Mover (automation with os, shutil)

The following pages describe each task in detail, including the approach taken, key challenges, and outcomes.

Task 1 - Hangman Game

Goal: Build a classic word-guessing game where the player must guess a hidden word one letter at a time before running out of attempts.

Approach:

- A list of words is defined; one is chosen at random using random.choice().
- The word is displayed as underscores, revealing matched letters.
- The player has 6 attempts; each wrong guess decrements the counter.
- Input is validated: single alphabetic character only (a-z).
- Guesses are case-insensitive (converted to lowercase).
- Repeated guesses are detected and rejected with a message.
- EOFError/KeyboardInterrupt are caught for clean exit.

Outcome: A fully functional terminal-based game with robust input handling and clear win/loss feedback.

Task 2 - Stock Portfolio Tracker

Goal: Create a CLI tool that calculates the total investment value of a stock portfolio based on user input and predefined stock prices.

Approach:
- A hardcoded dictionary maps tickers (AAPL, TSLA, MSFT, etc.) to prices.
- The user enters stock ticker and quantity in a loop (blank line to finish).
- Input validation checks for valid ticker, positive integer quantity.
- Per-stock value (price x quantity) and total are displayed in a table.
- The user can optionally export results to a .csv or .txt file.

Outcome: A practical tool demonstrating dictionaries, input validation, basic arithmetic, formatted console output, and optional file persistence.

Task 3 - JPG File Mover

Goal: Automate the repetitive task of collecting all .jpg files from a directory into a dedicated archive folder.

Approach:
- The user specifies source and destination folders (defaults provided).
- The destination folder is created automatically if it does not exist.
- The script scans the source directory using os.listdir().
- Files ending in .jpg (case-insensitive) are moved via shutil.move().
- Non-jpg files are left untouched.
- A summary reports how many files were moved.

Outcome: A simple but effective automation script saving manual effort when organising image files.

Conclusion

This internship provided a structured introduction to real-world Python application development. Each task reinforced foundational skills:

- Control flow, loops, and conditionals in game logic
- Data structures (lists, dictionaries, sets) for organising information
- Input/output and user interaction patterns
- File system operations and automation with the standard library

Beyond the technical skills, the experience demonstrated how to approach a problem, break it into manageable pieces, and iterate toward a working solution. The optional polish items for each task encouraged deeper thinking about edge cases and user experience.

Overall, the CodeAlpha internship served as a valuable stepping stone in Python development, building confidence to tackle more complex projects.